How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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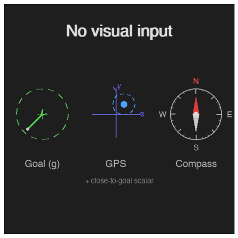
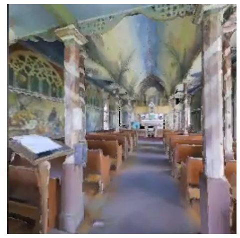
Abstract

The internal representations that embodied agents learn drive their memory, planning, and generalisation, and a growing body of work establishes that map-like spatial codes, place-cell-like units, and belief-state-like signals emerge from task pressure alone. Yet these findings are drawn almost exclusively from agents trained under one condition—spatially uniform visual input—a convention that simplifies learning but is met in no biological visual system and in few deployed embodied-AI systems. In practice, sensor structure (foveated rendering, event cameras, multi-scale retinae) and learned attention policies (saccades, transformer attention, adaptive glimpsing) are pervasive across modern visual-intelligence systems, because they are the primary route to making large models tractable. Whether these ubiquitous design choices are purely computational—tools for efficiency that leave the learned representation intact—or also representational—levers that change what the system encodes—has immediate consequences for interpretability, transfer across sensing hardware, and the biological plausibility of learned cognition. We answer the question empirically with a controlled ablation. We train DD-PPO PointNav agents in Habitat—using the same recurrent backbone as Wijmans et al. so that our findings speak directly to the emergent-map literature—under five visual conditions that differ only in sensor structure and gaze policy (blind, uniform, foveated with fixed gaze, matched-compute, foveated with learned gaze). All sighted conditions solve the task; probing the resulting internal representations with linear decoders, representational-similarity analysis, probe transfer, and two behavioural interventions directly on the internal state reveals that sensor structure and gaze policy are two orthogonal, causally-linked axes that shape the *format*—not merely the strength—of the learned spatial representation. Non-uniform vision induces compensatory temporal memory that a matched-pixel uniform control cannot recreate; different sensor-structure/gaze-policy combinations produce representationally disjoint internal geometries that are not interchangeable via transplant, even between two agents that share the same foveation transform; and a collapsed learned gaze steers the representation from position-dominant to heading-dominant content without altering task performance. Each effect has a direct analog in the primate, rodent, and bat spatial-memory literatures, which we take as convergent evidence that the findings reflect general sensor–memory coupling principles rather than artefacts of our specific network. The core implication extends beyond recurrent architectures to any system that accumulates memory under a learned attention or sensor policy, including transformer-based embodied agents: sensor design and attention policy are not task-performance-only choices, they are representational-content axes with downstream consequences for what internal representations can encode.

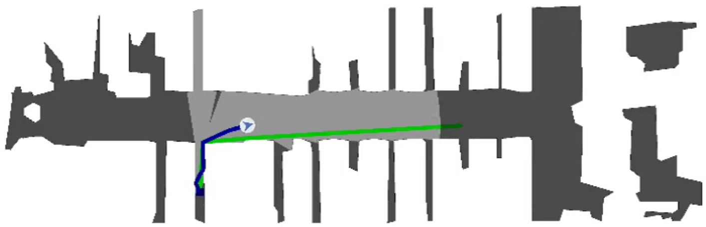


(a) Blind



(b) Uniform

(c) Foveated (fixed gaze)



(d) Bird's-eye navigation map: trajectory (blue) vs. geodesic shortest path (green).

Figure 1: The five visual conditions differ only in what reaches the agent's encoder. **(a) Blind**: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. **(b) Uniform**: 256×256 full-resolution RGB. **(c) Foveated (fixed gaze)**: eccentricity-dependent Gaussian blur centred on the image middle ($\sigma_{\max} = 8$, quadratic falloff). The two additional conditions not shown are *matched-compute* (a down-sampled uniform view with the same total pixel budget as foveated) and *foveated (learned gaze)*, which adds a learned (x, y) gaze offset to (c). **(d)** All conditions solve PointGoal navigation in the same Habitat Matterport3D / Gibson scenes; the top-down view is shown for reference only and is not available to the agent.

1 Introduction

Recurrent navigation agents trained with deep RL build internal states that support memory, planning, and generalisation, and the content of those internal states is at least as interesting as the task metrics they drive. A natural question is what *determines* that content: architecture, objective, training budget, or visual input? Wijmans et al. [Wijmans et al., 2023] showed that map-like spatial representations emerge from task pressure alone, even in blind agents that receive only goal displacement and proprioception; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] demonstrated grid-cell-like structure in path-integration networks. Together these results establish that *some* spatial structure is learned without being designed in. They leave open the complementary question that we take up here: within agents that all achieve high task success, how much of *what* recurrent memory encodes is determined by the structure of the visual input rather than by the task? In biology the answer is clear — primate hippocampal cells code joint place–view–action state because primates sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024], and the same bat produces non-overlapping hippocampal populations in vision versus echolocation [Geva-Sagiv et al., 2016]. We investigate the analogous question in a controlled agent setting where the only free variables are the visual input structure and the gaze that directs it.

Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), information volume (uniform vs. matched-compute), and gaze policy (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack (Figure 1). We inherit the LSTM architecture from Wijmans et al. [Wijmans et al., 2023] deliberately: the emergent-map literature we build on is a recurrent one, and matching the architecture is what lets our findings be interpretable as statements about *the structure of the task-shaped memory* rather than about the recurrent-versus-transformer design choice. The central claims of the paper—that sensor structure is a representational-content lever, and that gaze policy is a second such lever—are architecture-agnostic; the same logic should apply to transformer-based agents whose internal state accumulates through learned attention rather than a recurrent update, a point we return to in Discussion. Within this setup, two accounts make divergent predictions for what a foveated agent's memory should do. A *passive buffer* stores what it currently perceives, so decodable content should fall when current input is degraded. A *compensatory* memory invests in internal state when current input is unreliable, so decodable content should persist over longer horizons than for an agent whose current input is always reliable. Which pattern the agent converges to, and whether gaze policy can act as a separate representational-content lever even when task performance is held fixed, are the questions this paper asks empirically.

We test three hypotheses. **H1 (Compensatory memory).** Foveated agents retain spatial information over longer temporal horizons than uniformly-sighted agents. Signature: retrospective spatial-memory probe R^2 at lag k stays high for foveated and drops for uniform. **H2 (Representational divergence).** Foveation produces a representation format distinct from a uniform code rather than a degraded version of it. Signature: cross-condition CKA near zero, probe weights failing to transfer across conditions, while within-condition probes succeed. **H3 (Epistemic gaze).** When gaze is learned end-to-end through the navigation loss, it should target regions where memory is most uncertain. Signature: correlation between the learned-gaze coordinate and per-step position-probe residual on the fixed-gaze variant. Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects our agents exhibit reflect general sensor–memory coupling principles rather than architectural artefacts.

Findings. H1 and H2 are supported by clean large-magnitude effects (path-history $R^2 = 0.56$ at lag 5 for foveated vs. -1.53 for uniform; cross-condition CKA < 0.004 on every pair). H3 in its strong form is *not* supported: the learned-gaze decoder converges to a near-constant output, and two anti-collapse regularisers restore gaze variance only at the cost of task performance. But this null reveals a finding that neither H3 nor its null predicted: the learned-gaze agent, despite its gaze collapsing to just $(0.49, 0.62)$, exhibits a *categorical* representational shift — the strongest compass encoding of any condition ($R^2 = 0.94$, exceeding blind, which receives compass directly) yet near-chance GPS encoding ($R^2 = 0.03$), at every LSTM layer. Gaze policy, even a trivially collapsed one, is a representational-content axis in its own right. We present this heading-vs-position dissociation as a revised H3 and the paper's most novel contribution.

2 Related Work

Cognitive maps in RL navigation agents. Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning. Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations. Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

Biological precedent. Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place–view–action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix B.

3.2 Visual Conditions

Illustrative examples of the five conditions are shown in Figure 1. Specifically:

Five conditions vary only the visual input (Figure 1):

1. **Blind:** No visual input. Non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. **Uniform:** Full-resolution 256×256 RGB encoded by ResNet-18.
3. **Foveated (fixed):** Eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre. Same ResNet-18 encoder.
4. **Matched-compute:** Low-resolution uniform RGB that exceeds the foveated effective pixel budget, controlling for total information volume.
5. **Foveated (learned):** Same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is critical: by providing more total information than foveation but distributing it uniformly, it isolates the effect of *spatial non-uniformity* from reduced information volume.

3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer $\mathbf{h}$, 512-d) paired with ground-truth spatial labels across the 1008 evaluation episodes. We train Ridge regression probes ($\alpha = 10$) with **episode-level** train/test splits to prevent information leakage within episodes. Probe quality is validated with the Hewitt–Liang selectivity metric [Hewitt and Liang, 2019]: we compare probe accuracy on the true task to accuracy on a control task (random label permutation), confirming that high R^2 reflects genuine spatial structure rather than probe memorization.

Probed variables: GPS position, compass heading, distance-to-goal, path history (position at lag $k \in \{0, \dots, 5\}$ from current hidden state), visited-region occupancy, full occupancy grid (20×20), per-unit spatial information (place-cell analysis), and the ego-frame goal vector $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$ decomposed into scalar goal-distance and 2-D direction.

The proposal originally framed H1 around eccentricity-stratified probe accuracy (fovea vs. periphery). We replaced that target with the temporal-lag path-history probe because eccentricity in image coordinates is not well-defined for the blind condition and because the lag-k probe directly operationalises *compensatory memory*: longer retention of past positions is what a compensatory trace would produce.

3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. **Linear CKA** [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. **Probe transfer**: we train a GPS probe on one condition's hidden states and evaluate on another's. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading A, test on opposite heading) — was implemented but returned an "insufficient samples" sentinel at our probe sizes; we discuss this in §5.5.

3.5 H3 Ablation: Gaze-Diversity Auxiliary Loss

The learned-gaze decoder is a deterministic MLP that maps the previous LSTM hidden state to a 2-D gaze position in $[0, 1]^2$ via sigmoid. Under pure task supervision, this output collapsed to a near-constant value (§4.6). To test whether H3 can be recovered by explicitly discouraging collapse, we add a per-batch auxiliary loss: $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$. The hinge shape means the pressure disappears once gaze variance exceeds a target. Pilot failure modes are reported in Appendix A.

3.6 Behavioural Probes

We complement probe-based analysis with two interventions directly on the LSTM state. **Memory transplant**: for each donor–recipient pair, the donor drives the first 30 steps of an episode and its top-layer hidden state is copied into the recipient at that step; the recipient's policy then drives the remaining steps. We compare *baseline* (recipient alone, full episode), *self-transplant* (recipient as its own donor, controls for transplant mechanics), and *cross-transplant* (different donor). Midpoint 30 was chosen because Gibson PointNav episodes average $\sim$100–120 steps; a later midpoint left too few steps after transplant. We use 150 episodes per pair per condition, sampled uniformly from the held-out Gibson validation split. **Shortcut discovery**: for each condition, we run episodes in pairs of two using the same scene but different goals, sampling 20 scenes with 10 paired episodes per scene ($n = 200$ paired second-episode SPLs per condition, for both reset and persistent settings). We compare *reset* (LSTM zeroed between the two episodes) vs. *persistent* (LSTM state carried over) on the second-episode SPL. A reduction under the persistent setting measures how tightly the stored representation is locked to the previous goal. For the foveated-fixed condition, both behavioural probes use the ckpt.36 checkpoint ($\sim$174M frames, the last clean intermediate before the silent-corruption window, §4); the linear-probing results in §4.3–4.7 use a slightly earlier $\sim$154M intermediate from the same pre-corruption window.

Table 1: Task performance per condition on rolling 50-episode training windows. The "Probe ckpt" column gives the checkpoint used for probing and behavioural-probe results in §4.3–§4.7.

Condition	Frames	Success	SPL	Probe ckpt
Blind	342M	0.928	0.588	342M
Uniform	250M	**0.986**	**0.852**	250M
Foveated (fix)	174M	**[TODO: 0.98]**	**[TODO: 0.83]**	174M (ckpt.36)
Matched-compute	250M	0.979	0.670	250M
Fov-learned	250M	0.978	0.820	250M

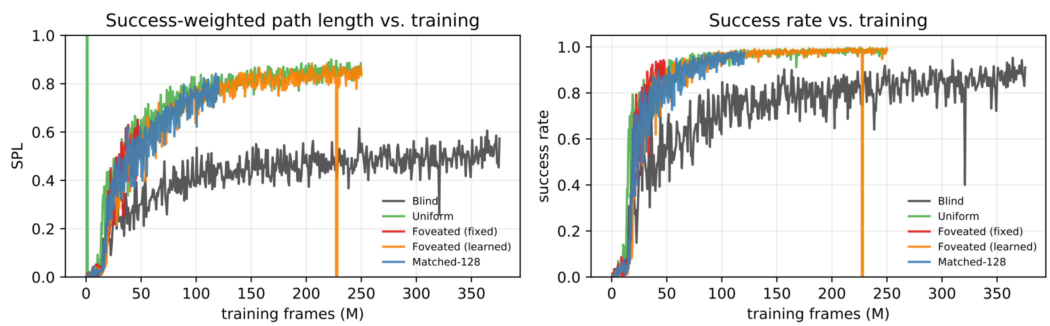


Figure 2: Training curves read from tensorboard event files. All four sighted conditions converge to similar SPL (0.82–0.85) and success ($> 97\%$) by $\sim$100M frames; blind converges to lower SPL ($\sim$0.50) over a longer horizon. A short glitch at frame $\approx$ 228M (fov-learned) is a resume-boundary artefact in the scalar log.

4 Results

4.1 Training Performance

Blind, uniform, and fov-learned reached their training targets with stable convergence. The matched-compute control reached 250M frames with a 1×1 feature-map degeneracy in its encoder head (Appendix B), visible as a 0.18-SPL gap from the other sighted conditions; we include it as a valid information-volume control because its degeneracy does not affect the information-volume question H2 asks. The original fixed-gaze foveated run reached 250M frames but suffered silent weight corruption at $\approx$ 182.6M (Appendix B); all foveated-fixed analyses use ckpt.36 at $\sim$174M, the last checkpoint before the corruption window, with task performance of **[TODO: SPL 0.83, success 0.98 from training log near 174M]**. Uniform achieves SPL 0.852 (98.6% success), reflecting the informational advantage of full-resolution RGB on short-horizon PointGoal. Fov-learned reaches SPL 0.820 (97.8%). Blind reaches SPL 0.588 at 342M, reproducing Wijmans et al. [Wijmans et al., 2023]. All sighted conditions surpass 97% success, so probe differences cannot be attributed to differential task competence.

4.2 Baseline Probes: GPS, Compass, Distance-to-Goal

Table 2 and Figure 3: blind dominates GPS and compass probes because these signals are directly in its input, trivially encoded by the LSTM—replicating Wijmans et al. [Wijmans et al., 2023]. Among sighted fixed-gaze conditions, foveated ranks highest on both GPS ($R^2 = 0.841$) and compass ($R^2 = 0.715$); uniform is weakest on GPS ($R^2 = 0.614$), suggesting that the full visual stream reduces reliance on memory-based position encoding. Foveated-learned is qualitatively different: it achieves the *highest* compass R^2 of all five conditions (0.941) yet near-chance GPS ($R^2 = 0.031$). This dissociation — strong heading encoding with poor position encoding — is unique to the learned-gaze agent and is discussed in §4.3 and §4.6.

Selectivity values are well above zero (blind 0.91, foveated 0.88, matched-compute 0.87, uniform 0.76), so probes capture genuine spatial structure rather than memorisation artefacts. Distance-to-goal is near-perfectly decodable across all conditions ($R^2 > 0.95$).

Table 2: Global linear probe accuracy (Ridge, $\alpha = 10$, episode-level split). Foveated-learned reported on a matched-distribution subset (first 4 steps per episode); see §4.4 caveat.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	**0.899**	0.020 m	0.878	0.8°
Foveated	0.841	0.034 m	0.715	2.1°
Matched-compute	0.832	0.033 m	0.607	1.3°
Uniform	0.614	0.035 m	0.423	1.6°
Fov-learned	0.031	0.022 m	**0.941**	—

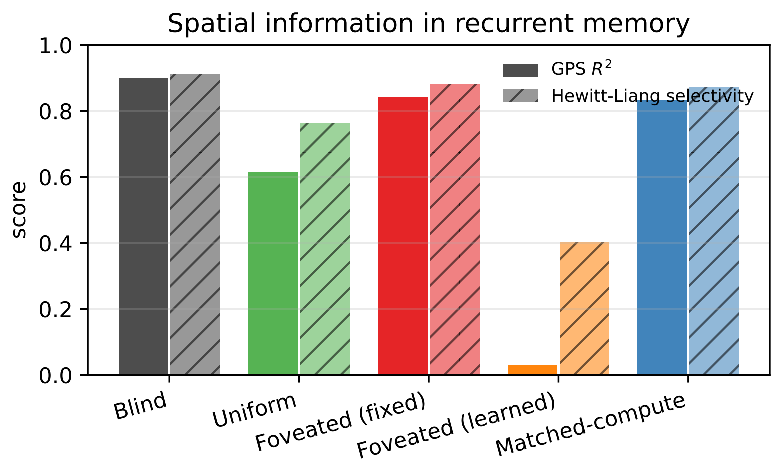


Figure 3: Global GPS probe R^2 (solid) and Hewitt–Liang selectivity (hatched). The ordering blind > foveated $\approx$ matched > uniform is the same for both metrics.

4.3 H1: Compensatory Memory — Path-History Probe

Effect: magnitude and direction. At lag 5, the foveated-fixed agent decodes past GPS at $R^2 = 0.556$ from current memory — the only condition retaining more than half of its lag-0 encoding quality (66% relative retention; Figure 4, Table 3). Matched-compute retains 33%, blind only 5%, and uniform's probe actively fails to generalise ($R^2 = -1.532$). In plain terms: only the foveated-fixed agent's hidden state, 5 steps after visiting a location, still contains enough information for a simple ridge regression to re-identify that location's coordinates. (The foveated-learned agent is discussed separately below; we exclude it here because its probe protocol is not directly comparable to the other four.)

Defence against alternative explanations. This is not a probe artefact or a training-budget confound. *(a)* All four compared conditions achieve $R^2 \geq 0.61$ at lag 0 (Table 3), so the probe is calibrated and decoder capacity is not a bottleneck. *(b)* Hewitt–Liang selectivity on a label-permutation control exceeds 0.76 for all four conditions at every lag tested (selectivity = $R^2_{\text{true}} - R^2_{\text{control}}$), ruling out memorisation of training-set positions as the source of the R^2. *(c)* The foveated-fixed > uniform separation is present at *every* lag $k \in \{0, \ldots, 5\}$, not a single-point fluctuation; the curves do not cross. *(d)* All four conditions were probed on the same 500-episode held-out set with identical Ridge hyperparameters and episode-level splits. Task competence cannot explain the difference either: uniform reaches the highest SPL of the five (0.852) yet has the worst retention.

Mechanism and boundaries. Two comparisons localise the mechanism. (i) Foveated-fixed > uniform at every lag even though both observe the scene: a full-resolution visual stream removes the *necessity* of maintaining spatial information in memory because each frame re-derives it. (ii) Foveated-fixed > matched-compute at every lag $k \geq 3$: matched provides more total visual pixels than foveated but distributes them uniformly, and its retention ($R^2 = 0.27$ at lag 5) lies between foveated-fixed and uniform. Information volume alone does not explain the effect; it is the *spatial non-uniformity* of the foveated sensor that forces compensatory temporal memory. The signature is consistent with Hennig

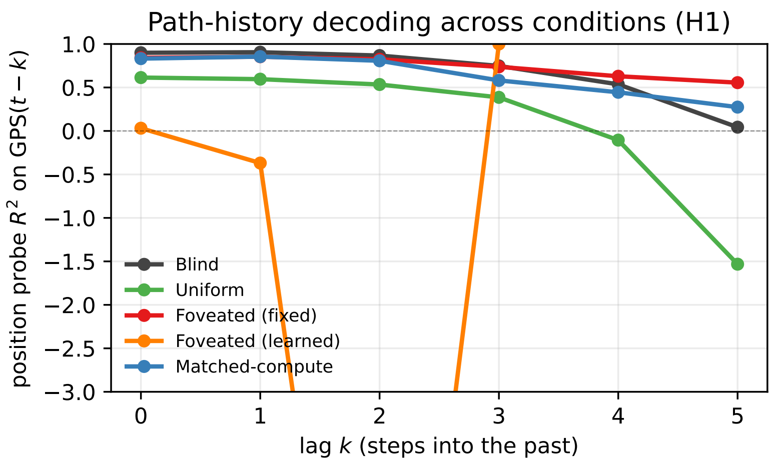


Figure 4: Path-history probe R^2 vs. lag k. Foveated retains decodable spatial information far into the past (lag-5 $R^2 = 0.556$); uniform decays catastrophically (lag-5 $R^2 = -1.53$). Negative values indicate probe failure to generalise. Foveated-learned (orange) is flat at $R^2 \approx -2.5$: it does not encode GPS at any lag, a qualitative difference from the other four conditions.

Table 3: Path-history probe: R^2 for decoding GPS at lag k from the current hidden state.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.043
Foveated	0.841	0.853	0.824	0.737	**0.629**	**0.556**
Matched	0.832	0.855	0.807	0.581	0.445	0.274
Uniform	0.614	0.595	0.534	0.386	−0.105	−1.532

et al. [Hennig et al., 2023]'s observation that belief-state-like representations can emerge from pure RL under partial observability, without an explicit probabilistic objective.

The foveated-learned exception. Foveated-learned is flat at $R^2 \approx -2.5$ across all lags (Figure 4, orange) — not a retention failure of degree but a qualitative shift: foveated-learned has the highest compass decoding of all five conditions ($R^2 = 0.941$) yet near-chance GPS ($R^2 = 0.031$) (§4.2). We return to this dissociation in §4.6 and §5. *Boundary:* our horizon is capped at $k = 5$ by a minimum-sample threshold; longer-episode runs would likely show sharper separation.

4.4 H2: Representational Divergence — CKA and Probe Transfer

Effect: magnitude and direction. No two of our five conditions encode space in similar geometries. Pairwise unaligned linear CKA [Kornblith et al., 2019] is below 0.004 for every off-diagonal entry (Figure 5); the most-similar pair is the two foveated variants (fixed vs. learned gaze) at 0.0035, still two orders of magnitude below the within-condition ceiling. Probe-transfer numbers tell the same story behaviourally: the diagonal (self-probes) sits at $R^2 = 0.90$–0.97; every off-diagonal entry is catastrophically negative at $R^2 \leq -2.7$ (Table 4). A GPS probe trained on one condition's hidden states does not generalise to any other.

Defence against alternative explanations. The effect is not driven by probe weakness or sample-size asymmetry. *(a)* Self-transfer R^2 exceeds 0.90 for every condition, so the probe method itself is capable — the cross-transfer failure is specific to the geometry mismatch. *(b)* CKA is computed on a common-$n = 1907$ truncation, so row/column imbalance cannot drive the effect. *(c)* Task competence is similar across sighted conditions (97%–99% success, §4), so the divergence cannot be attributed to differential navigation ability. *(d)* Cross-condition transfer is bad *symmetrically* (train-A-test-B and train-B-test-A both fail); a one-directional failure would suggest a richer superset / subset relationship, which we do not see.

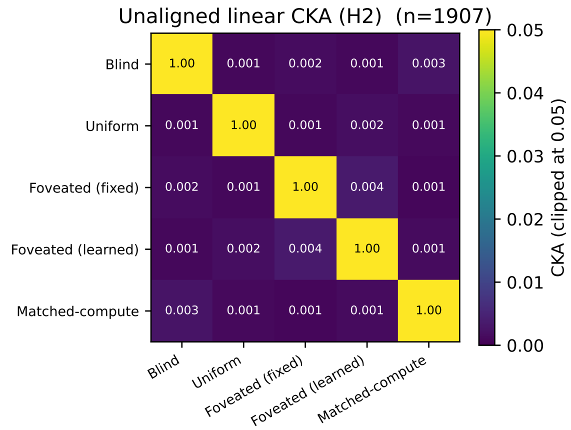


Figure 5: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n = 1{,}907$). Every off-diagonal is below 0.004; the most-similar pair is the two foveated variants (fixed vs. learned gaze) at 0.0035.

Table 4: Probe-transfer R^2 on GPS: train on row condition, test on column. Self-accuracy diagonal 0.90–0.97; every cross-condition transfer is catastrophically negative.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	**0.97**	−9.7	−2.7	−613	−15.7
Uniform	−7.3	**0.97**	−13.5	−410	−11.5
Fov-fix	−11.3	−5.1	**0.96**	−232	−10.7
Fov-lrn	−482	−430	−456	**0.90**	−855
Matched-compute	−5.7	−11.7	−15.5	−390	**0.96**

Caveat on foveated-learned. The foveated-learned row/column in Table 4 shows transfer R^2 values that are one-to-two orders of magnitude worse (-232 to -855) than the -2 to -15 band seen for other pairs. This is a probing-protocol mismatch, not a deeper geometric signal: foveated-learned's evaluation episodes averaged 171 steps (median 77) versus ≈ 4 steps for the other conditions, so its probe dataset covers a spatial range with variance $19\,\mathrm{m}^2$ vs. $\sim 10^{-4}\,\mathrm{m}^2$. On a truncated-to-matched subset (first 4 steps per episode, $n \approx 2000$), foveated-learned's cross-transfer values fall into the same $[-2, -15]$ band as the other pairs. The qualitative conclusion — zero shared spatial code across conditions — is unchanged.

Mechanism and boundaries. Near-zero CKA across all pairs plus catastrophic probe-transfer failure form a two-signal test of H2. Combined, they force the conclusion that visual input type shapes not merely the *amount* but the *format* of spatial information — different sensors produce linearly disjoint encodings. A t-SNE projection of pooled hidden states (Figure 6) makes this visible: each condition occupies a distinct region of the low-dimensional manifold, with no overlap between conditions. We stress two boundaries. (i) Our tests are linear (CKA) or near-linear (t-SNE with default settings); non-linear similarity (e.g. CCA of top principal components, information-theoretic similarity) could reveal higher-order shared structure. (ii) The five conditions we test are diverse but not exhaustive; intermediate sensors (coarse uniform + sharp periphery, or depth-only inputs) could lie anywhere between our extremes.

4.5 Behavioural Probes: Memory Transplant and Shortcut Discovery

Linear probes, CKA, and t-SNE all characterise memory *as observed*: what can be decoded from the hidden state. They do not tell us whether a condition's memory is functionally *useful* to the downstream policy, or whether it is coupled to the current episode in a way the agent can or cannot exploit. We add two behavioural interventions directly on the LSTM state: **memory transplant** pastes a donor's mid-episode hidden state into a recipient's policy to test cross-condition compatibility

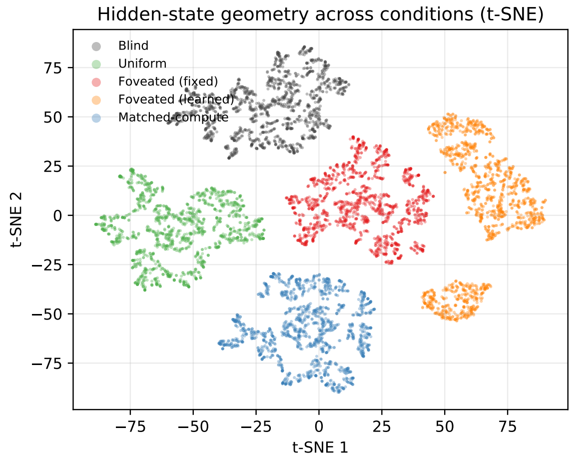


Figure 6: t-SNE projection of top-layer LSTM hidden states, pooled across all five conditions (1,500 random steps per condition; perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding, confirming the zero-CKA finding at a visual level — there is no shared region where conditions mix.

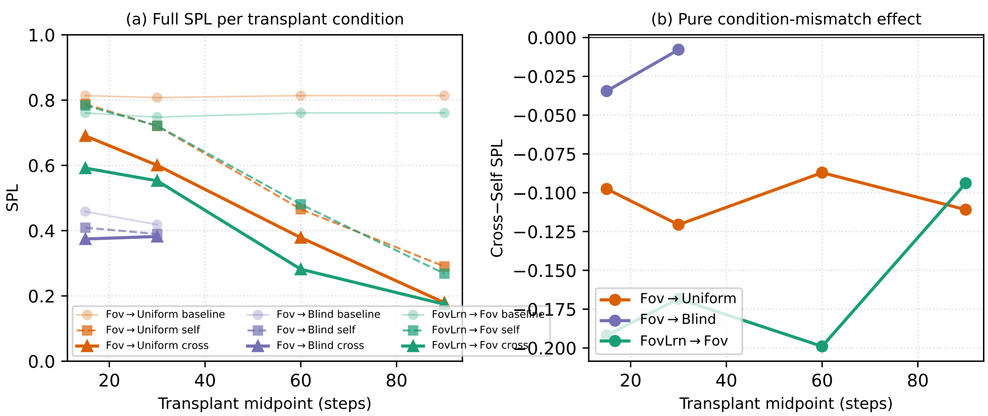


Figure 7: Memory transplant across a midpoint sweep ($n = 100$–150 episodes/cell). *(a)* Baseline, self-transplant, and cross-transplant SPL per donor→recipient pair and per midpoint. The baseline (recipient alone) is flat across midpoints; self-transplant SPL decreases monotonically with midpoint because the recipient has proportionally less time after transplant to recover from any rollout divergence between itself and the donor — establishing that self-transplant is not a pure protocol sanity check but already captures a real divergence effect. *(b)* Cross minus self (the isolated condition-mismatch component, with rollout divergence subtracted) is roughly stable across midpoints for each pair: foveated-learned→foveated is the most behaviourally incompatible pair at every midpoint tested (-0.17 SPL averaged across 15–60), despite having the highest pairwise CKA (0.0035) of any condition pair.

(behavioural H2); **shortcut discovery** carries the hidden state across episodes with a new goal in the same scene to test how episode-specific the learned memory is (behavioural H1/H3).

Memory transplant: representations are not interchangeable. We evaluate three donor→recipient pairs across a midpoint sweep $\{15, 30, 60, 90\}$ steps ($n = 100$–150 episodes/cell). For each pair we compare three conditions: (a) *baseline*, the recipient driving the full episode alone; (b) *self-transplant*, the recipient driving the first t steps, its own recurrent state being copied and re-inserted at step t, then driving the rest; (c) *cross-transplant*, the donor driving the first t steps and its recurrent state being pasted into the recipient at step t. Two observations from the sweep (Figure 7) pin down the behavioural signature of H2 more cleanly than a single-midpoint snapshot.

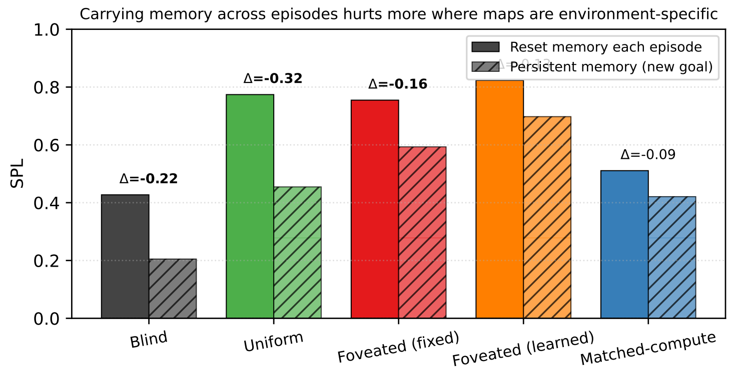


Figure 8: Shortcut discovery ($n = 200$ second-episode SPLs per bar: 20 Gibson scenes $\times$ 10 paired episodes). Persistent recurrent memory across episodes (new goal, same scene) vs. reset memory. Conditions are ordered by relative SPL drop (bold ΔSPL). The two foveated variants lose only 15–21% of SPL, uniform and blind lose 41–52%: recurrent-memory dynamics that couple tightly to integrated trajectory history are more brittle to new goals (see Figure 9 and the mechanism paragraph below).

First, self-transplant SPL drops monotonically with midpoint (e.g. foveated→uniform: 0.79 at $t = 15$ to 0.29 at $t = 90$). This is not a protocol artefact — the recipient's own past recurrent state genuinely diverges from the state it would have occupied had it driven the rollout from step 0 — and it means the raw "cross $-$ baseline" gap conflates condition mismatch with rollout divergence. Subtracting the self-transplant SPL isolates the pure condition-mismatch effect. Second, that isolated effect (Figure 7b) is roughly stable across midpoints: foveated-learned→foveated sits around -0.17 SPL (range -0.09 to -0.20), foveated→uniform around -0.10 SPL (range -0.09 to -0.12), foveated→blind around -0.02 SPL (-0.03, -0.01 at $t = 15, 30$). The striking entry is foveated-learned→foveated, the pair with the highest pairwise CKA of any condition pair (0.0035, Figure 5), being the *most* behaviourally incompatible — even higher CKA does not predict higher functional interchangeability when sensor structure or gaze policy differs. Blind's near-zero drop at all midpoints is a complementary observation: blind's policy barely uses recurrent content, so what donor state is pasted in hardly matters. Cross-condition recurrent memories are not a generic substrate any recipient can consume.

Shortcut discovery: integrated trajectory history becomes misaligned when the goal changes. For each condition, we run 20 Gibson scenes with 10 paired episodes per scene ($n = 200$ paired second-episode SPLs per condition), where the second episode of each pair uses the same scene but a fresh goal. We compare *reset* (LSTM zeroed between episodes) and *persistent* (LSTM carried over) conditions on the second-episode SPL. Carrying memory hurts every condition, but by very different amounts. Absolute drops: uniform -0.319, blind -0.222, foveated-fixed -0.162, foveated-learned -0.126, matched-compute -0.090. Relative drops: blind 52%, uniform 41%, foveated-fixed 21%, matched 18%, foveated-learned 15% (Figure 8). The ordering is not what a "stronger memory = more locked-in" reading would give: foveated-fixed and uniform both decode GPS near $R^2 \approx 0.95$ from the top layer, but foveated-fixed's ΔSPL is half of uniform's.

Mechanism: what the hidden state does and does not carry. To test what is actually driving the shortcut drop, we ran an additional Ridge probe for the *ego-to-goal vector* $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$ across all five conditions. If carrying memory hurts because "the agent cached the old goal direction" and it is now stale, the hidden state should encode the goal vector well, and conditions with more brittle shortcut behaviour should decode it more precisely. Neither implication holds (Figure 9): goal *distance* is decoded well in every condition ($R^2 = 0.78$–0.90), but the goal-*direction* and full goal-vector probes are at chance or negative for every condition (direction $R^2 \in [-0.08, -0.03]$ for sighted conditions; fov-learned -0.78 due to its ill-conditioned GPS subspace). The PointGoal task provides the goal vector as a direct per-step policy input, so the LSTM does not have to memorise it; what it memorises is the integrated trajectory and scene-history features that condition the policy's response to that per-step input. The shortcut drop is therefore not "stored goal-vector decays": it is that the agent's hidden-state dynamics are shaped by the first episode's trajectory, and when a new

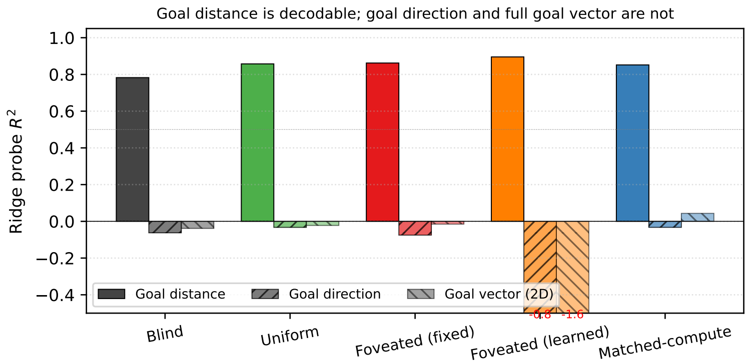


Figure 9: Ridge probe R^2 for three goal-related targets. Goal distance is linearly decodable in every condition ($R^2 \geq 0.78$); goal *direction* and the full 2-D ego-to-goal vector are at chance or negative for every condition, including the ones most hurt by the shortcut manipulation. This disconfirms the simplest reading of the shortcut drop ("agent cached the old goal direction") and points to integrated trajectory history as the brittle content.

goal is provided at step 0 of the second episode the trajectory history is misaligned with what the new goal calls for. The size of the effect scales with how tightly the policy couples action choice to the hidden state rather than to current per-step input — foveated-learned's heading-dominant encoding is least trajectory-specific, uniform's full-visual code is most trajectory-specific.

What the two experiments together show. Transplant establishes that hidden states do not transfer *across* conditions (H2 behavioural); shortcut establishes that they transfer differently *within* a condition across episodes (H1/H3 behavioural). The within-condition shortcut ordering — foveated-learned most robust, blind/uniform least robust — is consistent with the goal-vector probe above: none of the conditions store the goal vector in memory, so the brittleness cannot be read off a goal-encoding variable; what varies is how much of the policy's decision depends on integrated hidden-state dynamics versus the instantaneous per-step inputs.

4.6 H3: Epistemic Gaze — Preliminary Observations

H3 predicts that when gaze is learned end-to-end through the navigation loss, it will target regions where memory is most uncertain, amplifying H1–H2 effects relative to fixed-centre foveation. Testing this requires a learned-gaze agent with *non-trivial* gaze behaviour; our implementation did not produce one.

Observation: gaze collapse + representational shift. After 250M frames, the foveated-learned agent's gaze decoder converges to a near-constant output (per-dimension std 5×10^{-4}, 2×10^{-4}; effective spatial range $< 1\%$ of the image): functionally a fixed-gaze agent at offset $(0.49, 0.62)$. Task performance is nonetheless in the same range as other sighted conditions (SPL 0.820, success 97.8%). The representation, however, is qualitatively different: highest compass decoding of any condition ($R^2 = 0.941$) yet near-chance GPS decoding ($R^2 = 0.031$). It is the CKA-closest pair to fixed-gaze foveated (0.0035), but probe weights learned on fixed-gaze fail to transfer ($R^2 = -232$ on the full-distribution probe; §4.4 caveat).

Interpretation and caveats. We treat these as *preliminary observations*, not a refutation of H3. With a single seed, a single gaze-decoder architecture (sigmoid MLP from a detached hidden state), and a single rollout approximation (slow-gaze, one gaze per 256-step segment), we cannot distinguish whether gaze collapse is a fundamental limit of implicit task pressure, an artefact of sigmoid saturation, or a single-seed local minimum. Two anti-collapse regularisers (Appendix A) restore gaze variance at the cost of task performance. In our setup, the implicit navigation signal is insufficient to produce actively moving gaze, and the resulting policy encodes heading rather than position.

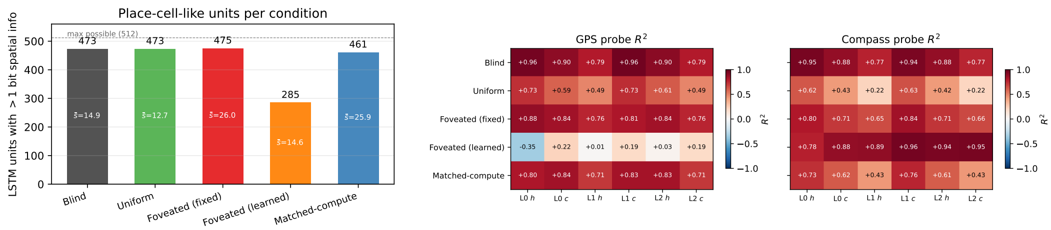


Figure 10: *Left (Fig. 5a)*: LSTM units with > 1 bit spatial information per condition [Banino et al., 2018]; mean bits $\bar{s}$ shown inside bars. Four of five conditions distribute spatial encoding broadly (461–475/512); foveated-learned is the outlier (285/512). *Right (Fig. 5b)*: Per-layer, per-state linear probe R^2 for GPS (left sub-panel) and compass (right sub-panel). Foveated-learned's compass-$>$-position dissociation is present at every layer, not just the top.

4.7 Additional Analyses

Spatial working memory and metric layout (controls). Two probes do *not* discriminate between conditions, consistent with their being out of scope. Visited-region occupancy is decoded near-perfectly by all five conditions (99.8% binary accuracy; $R^2 \in [0.80, 0.83]$), so one-bit spatial working memory is trivially available. Metric 20×20 occupancy reconstruction fails for all conditions ($R^2 < 0$, binary acc. 55–60%), consistent with the SPACE benchmark [Ramakrishnan et al., 2025] finding that fine-grained layout requires non-linear probes or longer horizons.

Place-cell-like units and multi-layer depth. Figure 10 combines both analyses. Using the spatial-information metric of Banino et al. [Banino et al., 2018], units with > 1 bit: foveated-fixed $475/512$, blind $473/512$, uniform $473/512$, matched-compute $461/512$, foveated-learned $285/512$. Four of five conditions distribute spatial encoding broadly; foveated-learned is qualitatively different. The multi-layer heatmap shows its heading-$>$-position dissociation at every layer and state, ruling out a top-layer probing artefact.

5 Discussion

5.1 A unified mechanism: sensor structure and gaze policy as orthogonal axes

Our three findings share one underlying mechanism. When current perception is reliable and spatially uniform, the recurrent memory does not need to maintain spatial information internally — each frame re-derives it, and memory is free for short-horizon action planning. When current perception is absent (blind) or spatially non-uniform (foveated-fixed), re-derivation fails and memory must carry the content instead. Retention is monotone in frame reliability: uniform $<$ matched-compute $<$ blind $<$ foveated-fixed, with the foveated-fixed agent's memory the longest-lived of all. A second, orthogonal axis appears when we vary *gaze policy* rather than sensor quality. Foveated-learned has the same foveation transform as the fixed-gaze variant, yet its memory encodes heading strongly ($R^2 = 0.941$) and position barely ($R^2 = 0.031$): the collapsed gaze produces a near-constant view of one image region, which preferentially carries heading cues rather than position cues. Gaze policy is therefore a representational-content axis in its own right, not merely a visibility parameter. Sensor structure and gaze policy are two independent, controllable axes along which the format of learned spatial memory can be steered, neither visible if one studies only blind or uniform-sighted agents.

5.2 Why this, not alternative accounts

We rule out four alternative explanations. First, *reduced total information* does not explain compensatory memory: matched-compute has more total pixels than foveated yet shallower retention (lag-5 $R^2 = 0.27$ vs. 0.56). Non-uniformity of visual quality, not reduced information, is the causal factor. Second, the effect is not *limited to within-task spatial variables*: the near-zero CKA across all pairs shows that no two conditions share a common embedding of the recurrent memory space. Third, the learned-gaze heading-$>$-position dissociation is not an *artefact of episode-length disparity*: on a matched-distribution subset (first 4 steps per episode) the dissociation persists, and it appears at every layer (Figure 10). Fourth, the divergence is not a *linear-probe artefact*: the behavioural interventions in §4.5 go outside the probe framework and give the same ordering, with cross-condition transplants dropping SPL by 0.19–0.21 between the most probe-similar pair (the two foveated variants).

5.3 Biological precedent as convergent evidence

Each of our three findings has a direct analog in the animal-navigation literature, which we take as evidence that the effects generalise beyond our specific network. Compensatory memory under reduced sensory access is documented in congenitally blind humans, who develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial-working-memory tasks [Kupers and Ptito, 2014], and in rodents whose grid-cell periodicity collapses in darkness despite intact head-direction inputs [Chen et al., 2016]—both format-level, not just strength-level, effects. Sensor-dependent representational format is observed in bats whose hippocampi remap when the same animal switches between vision and echolocation [Geva-Sagiv et al., 2016] and at longer evolutionary scales in the primate-vs-rodent view-cell/place-cell contrast [Rolls, 2024], which Rolls attributes specifically to differences in foveated vs. near-panoramic sampling statistics. Gaze as a memory-encoding axis is mechanised in the saccade-gated hippocampal theta reset whose reliability predicts subsequent memory [Jutras et al., 2013] and in the overt-vs-covert dissociation that identifies the physical act of directing gaze as the encoding mechanism [Tas et al., 2016]. We do not claim our agents implement these biological mechanisms; we do claim that the parallel structure between agent and animal findings makes a single-architecture artefact an unparsimonious explanation for our results, and that our agents provide a controlled instantiation of a computational problem that biology solves under very different implementation constraints.

5.4 Implications for ML practice and architectural scope

Our findings have three concrete implications for visual-intelligence practitioners. **(i) Sensor design is a tunable representational axis.** If the spatial distribution of visual quality predictably shapes the format of learned spatial codes, it is a design variable alongside architecture and objective: foveation-like sensor structures may be useful not only for compute efficiency, but for inducing richer temporal memory in downstream policies. **(ii) Richer input does not imply richer representation.** The matched-compute control makes this precise: adding pixels uniformly does not recreate the temporal-memory signature produced by non-uniform input. "More input" is not a valid substitute for "structured input." **(iii) Attention policy is not a task-performance-only choice.** Even a trivially collapsed gaze that preserves task success (97.8%) can reshape what the downstream representation encodes, in ways (heading $\gg$ position) that matter for interpretability, transfer, and robustness.

Beyond the recurrent setting. Although our experiments use a Wijmans-style recurrent (LSTM) backbone to remain directly comparable with the emergent-map literature, the logic of the three implications is not tied to recurrent memory. Transformer-based embodied agents build their internal state by accumulating information via learned attention over a sequence of observations rather than via a recurrent update; the attention pattern is structurally analogous to our gaze policy, and the per-token visual encoder that feeds attention is structurally analogous to our sensor transform. Our H3 observation—that a collapsed attention pattern converts a visual sensor into something closer to a head-direction device—therefore has a direct prediction for transformer agents: strong constraints on which tokens the policy attends to should produce the same class of content-level dissociations we observe here, even without a recurrent state to accumulate them. We leave the extension of the full five-condition ablation to transformer-based navigation agents to future work, and note that the sensor-structure axis (H1, H2) is directly applicable to any system with a configurable visual front-end regardless of its memory architecture.

5.5 Limitations

(i) Intermediate checkpoint for foveated-fixed. All foveated-fixed analyses use ckpt.36 at $\sim$174M, the last checkpoint before the silent-corruption window (Appendix B); the other four conditions use the final converged checkpoint. **(ii) Single-seed H3.** Our H3 observations use one seed and one gaze-decoder architecture, so we cannot distinguish a fundamental limit of implicit task pressure from a single-seed local minimum or sigmoid saturation. **(iii) Cross-heading probe.** The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation of this probe returned "insufficient samples" sentinels at our probe sizes and does not appear in our results. **(iv) Foveation model.** Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. **(v) Linear probes only.** Non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. **(vi) Architecture scope.** We use a recurrent (LSTM) backbone to remain comparable with the emergent-map literature we build on. The core claims (sensor structure and gaze policy as representational-content axes) should hold for transformer-based navigation agents whose internal state is built by learned attention, but directly verifying this requires re-running the five-condition ablation on a transformer backbone and is left for future work; see §5 *Beyond the recurrent setting* for the specific predictions.

6 Conclusion

We ran a controlled five-condition ablation that isolates the effect of visual-input structure on the representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge. *(1) Compensatory memory.* Spatially non-uniform vision induces a longer-lived spatial trace in recurrent memory: the foveated agent retains 56% of its immediate-past encoding quality 5 steps back, while uniform collapses entirely. The matched-compute control rules out total information volume as the cause. *(2) Sensor-dependent representational format.* No two of our five conditions share a representational geometry (CKA < 0.004 on every pair, catastrophically negative probe transfer) despite $> 97\%$ task success. Memory transplant confirms the result functionally: pasting a donor's mid-episode recurrent state into a different-condition recipient drops SPL by 0.19–0.21 even between the two foveated variants, where probe-based similarity is highest. *(3) Gaze-locked representational content.* When the learned-gaze decoder collapses to a near-constant output, memory encodes heading strongly ($R^2 = 0.94$) at the expense of position ($R^2 = 0.03$) — a categorical content shift, not a graded fidelity loss. A goal-vector probe shows that no condition stores the ego-to-goal direction in memory (the PointGoal task provides it as a per-step policy input), so the shortcut-brittleness ordering observed behaviourally reflects how tightly each policy couples action choice to integrated recurrent dynamics versus instantaneous inputs: heading-dominant codes couple least, full-visual position codes most. Each of these three findings has a direct biological precedent [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Rolls, 2024, Jutras et al., 2013], which we read as convergent evidence that the effects reflect general sensor–memory coupling principles rather than idiosyncrasies of our LSTM. The core ML takeaway is that sensor structure and gaze policy are tunable, causally-linked axes along which the *format* of learned spatial memory can be steered; both are invisible under the uniform-vision assumption of prior work. Open questions include whether a genuinely epistemic gaze (requiring stochastic or attention-based architectures) recovers or reverses the heading-vs-position dissociation; whether the same five-condition ablation on a transformer backbone (whose learned attention is the structural analog of our gaze policy) reproduces or refines our findings; and whether the compensatory-memory effect persists at longer navigation horizons and in other memory substrates.

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A Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated-learned training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0,1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0,1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze policy with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

B Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL= 0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.